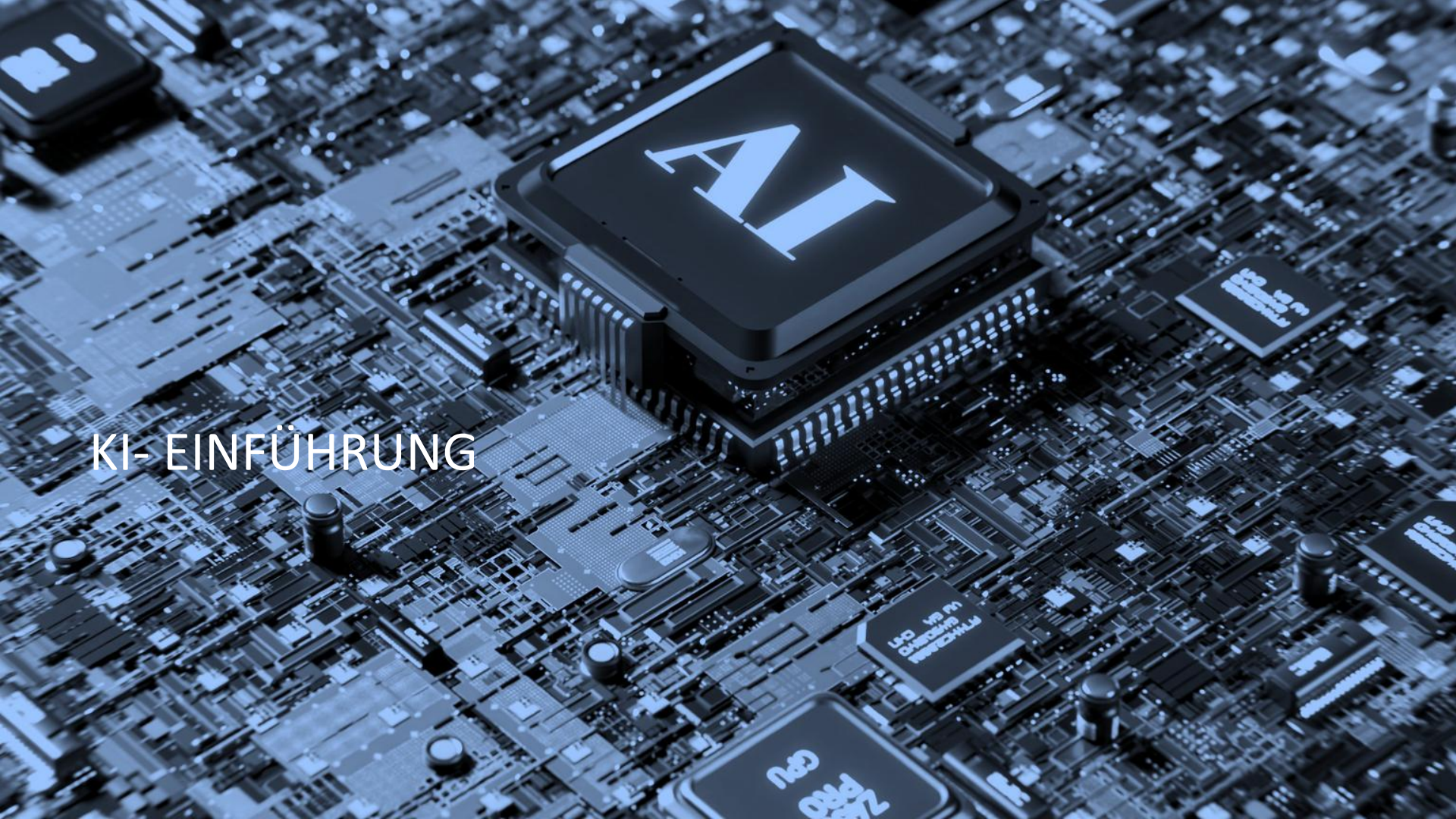




KI- EINFÜHRUNG

AGENDA

1. Tag: 09:30 – 14:00 Uhr

01

Künstliche Intelligenz

- Was ist KI?
- Wie funktioniert KI?
- Was kann KI?

3. Tag: 09:00 – 15:00 Uhr

03

Strategischer Umgang mit KI

- Erfassen des Status Quo
- Herausforderungen entdecken
- Lösungen implementieren

1. Tag: 14:00 – 17:00 Uhr
2. Tag: 09:00 – 17:00 Uhr

02

Rechtlicher Rahmen

- Grundlage
- KI EU-VO
- Compliance
- Governance

3. Tag: 15:00 – 17:00 Uhr

04

Individuelle Themen

- Offene Punkte
- Fragen
- Diskussion

KÜNSTLICHE INTELLIGENZ

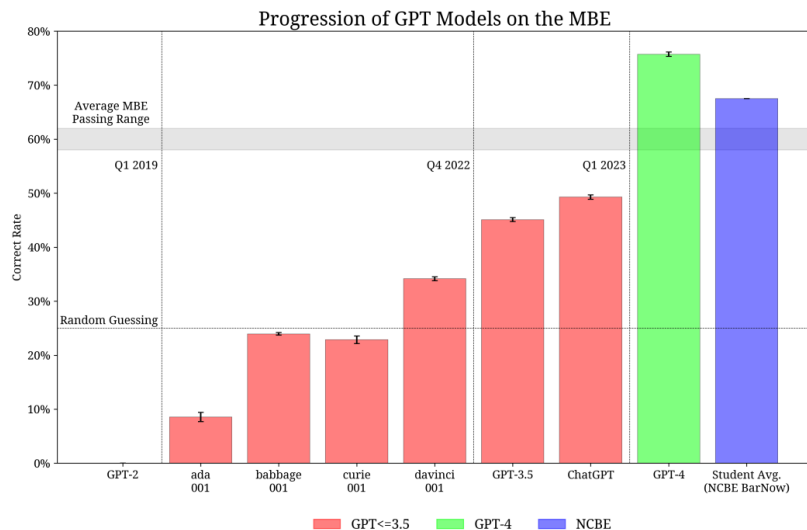
Verstehen



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KÜNSTLICHE INTELLIGENZ

BETTER THAN THE AVERAGE STUDENT



GPT-4 Passes the Bar Exam: Katz, Bommarito, Gao, 2023

<https://www.br.de/nachrichten/netzwelt/chatgpt-ki-besteht-bayerisches-abitur-mit-bravour,TfB3QBw>

KI-Abitur:

“So viel besser war GPT4”

Fach:	Abi 2022 / ChatGPT 3.5	Abi 2023 / ChatGPT 4
Deutsch	5 +	2 -
Mathematik	4 -	2-
Informatik	5	2
Ethik	4 -	2 -
Geschichte	3 +	2

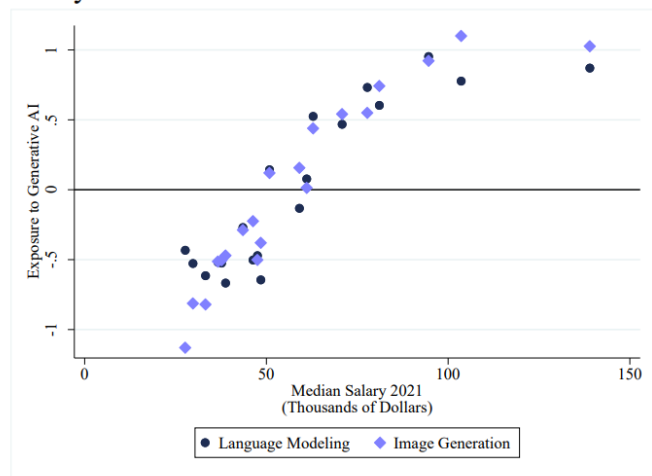


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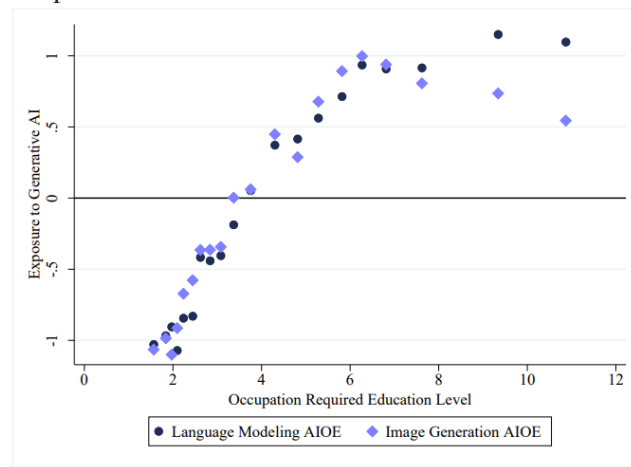
KÜNSTLICHE INTELLIGENZ

UNEXPECTEDLY AFFECTED OCCUPATIONS IN THE AGE OF AI

Panel A. Exposure to Generative AI and Median Salary.



Panel B. Exposure to Generative AI and Required Level of Education.



Occupational Heterogeneity in Exposure to Generative AI, Felten, Edward W. and Raj, Manav and Seamans, Robert, 2023: <http://dx.doi.org/10.2139/ssrn.4414065>



KÜNSTLICHE INTELLIGENZ

Beispiele von KI-Anwendungen in verschiedenen Anwendungsbereichen

Welche
Anwendungsfälle
kennen Sie?

Anwendungsbereich	Anwendungsfälle
Datenschutz/ -sicherheit	Spam-Filter von Mailanbietern, Kontrollmechanismen in sozialen Medien zur Löschung fragwürdiger Inhalte
Haussteuerung	Lernendes Thermostat zur individuellen Steuerung der Heizung
Logistik	Routenoptimierung, Lagerraumoptimierung, Zwischenlagerung von Paketen und Waren
Medien	Textgenerierung z. B. zu Sportereignissen, Börsennachrichten, Wetterberichten, Textübersetzung
Kunst	Kreierung von Bildern basierend auf Textvorgaben
Autonomes Fahren	Erkennung von Gefahrensituationen
Landwirtschaft	Teilautonom fahrende Traktoren und Mähmaschinen, vollautomatisierte Produktionsanlagen für Gemüse, Obst und Salat
Navigation	Berücksichtigung temporärer Verkehrszeichen, Baustellen und unvorhersehbarer Ereignisse wie Unfälle oder Wetterbeeinträchtigungen bei der Wegberechnung
Medizin	Bewertung von pathologischen Befunden, Bilddiagnose – Früherkennung
Umwelt	Vorhersage von Ernteerträgen, Pflanzenwachstum oder Umweltkatastrophen basierend auf Satellitenaufnahmen der Erde
Marketing	Anzeige nutzerspezifischer Angebote und Inhalte

KÜNSTLICHE INTELLIGENZ

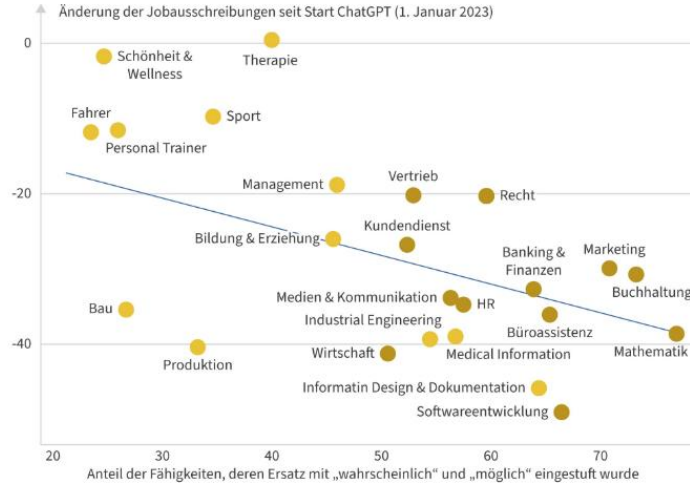
Welche Jobs sind betroffen?

Übung

Verhältnis zwischen GenAI-Exposure und Jobausschreibungen

Angaben in Prozent

● Berufe mit GenAI-Exposure > 50 Prozent ● Berufe mit GenAI-Exposure < 50 Prozent

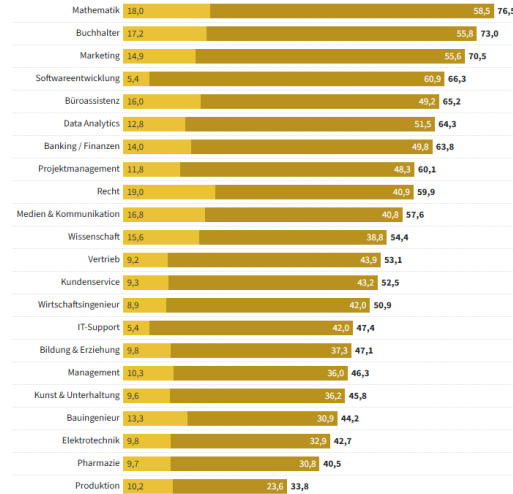


Grafik: saco. / Quelle: Indeed 2024

Jobs, die von ChatGPT ersetzt werden

In Stellenausschreibungen geforderte Fähigkeiten, die ChatGPT

● wahrscheinlich ersetzt ● möglicherweise ersetzt



ChatGPT-Exposure > 50 Prozent
Grafik: saco. / Quelle: Indeed 2024

<https://www.faz.net/pro/digitalwirtschaft/kuenstliche-intelligenz/die-jobs-die-von-der-ki-getroffen-werden-110062161.html>

KÜNSTLICHE INTELLIGENZ

UNEXPECTEDLY AFFECTED OCCUPATIONS IN THE AGE OF AI



„Once a new technology rolls over you, if you're not part of the steamroller, you're part of the road“ (Stewart Brand, American writer)

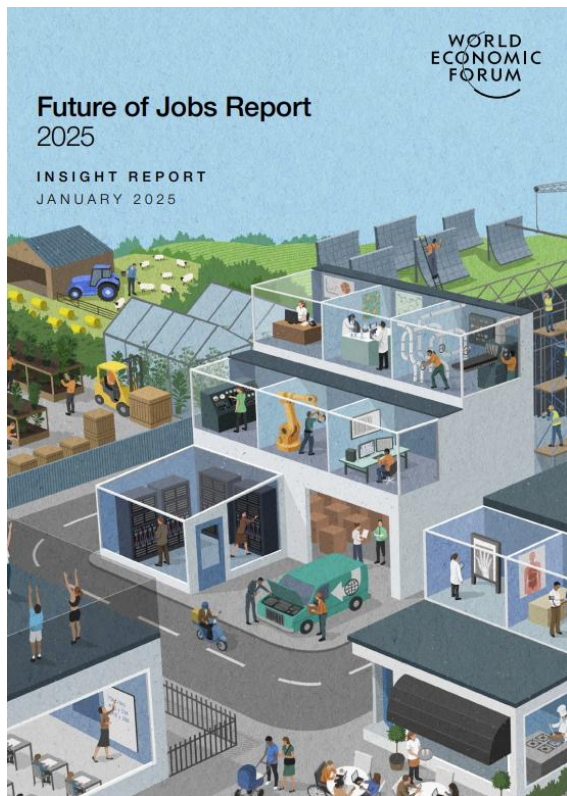


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KÜNSTLICHE INTELLIGENZ

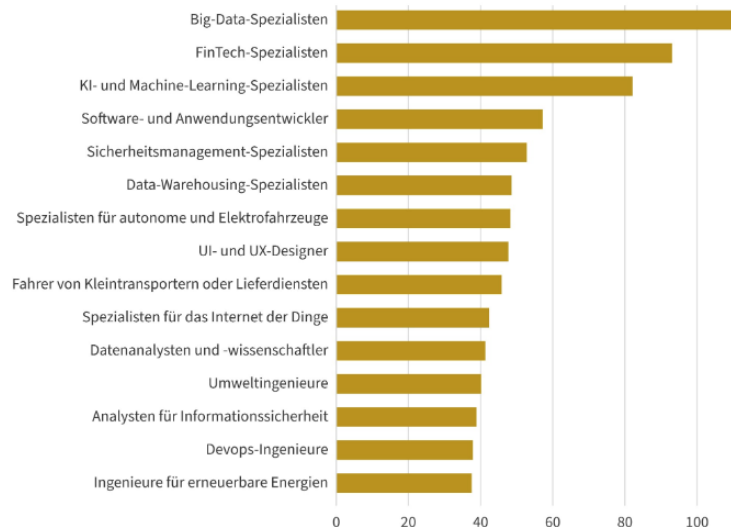
UNEXPECTEDLY AFFECTED OCCUPATIONS IN THE AGE OF AI

Übung



Job-Wachstum 2025 bis 2030

Erwarteter Zuwachs in Prozent der vorhandenen Arbeitsplätze



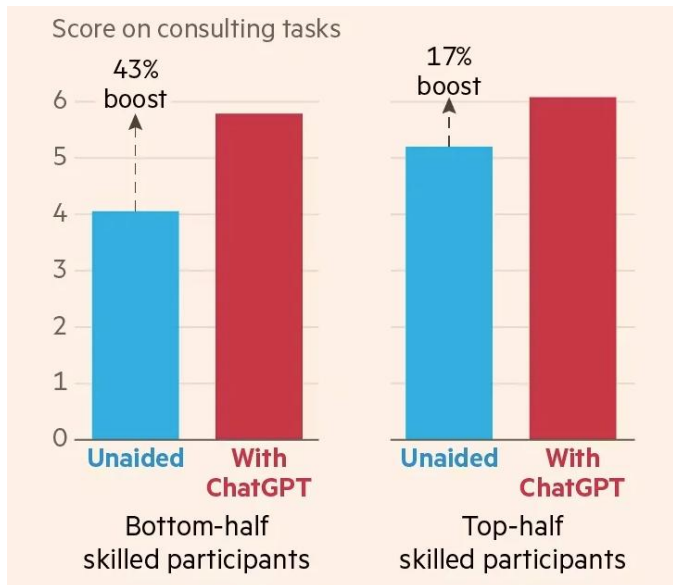
WEF Future of Jobs Survey
2025



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KÜNSTLICHE INTELLIGENZ

PERFORMANCE BOOST SCORE ON CONSULTING TASKS



Künstlich intelligent



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Creation of computer programs to imitate human behavior and abilities.

Machine Learning

Creation of computer programs to imitate human behavior and abilities that learn from experiences and data instead of being explicitly programmed.

Unsupervised Learning

Learning to recognize patterns without predefined outputs.

Supervised Learning

Learning from labeled data to make accurate predictions for new data.

Reinforcement Learning

Learning through trial and error to maximize rewards.

Deep Learning

Creation of artificial neural networks that focus on processing large amounts of data.

NLP

Approach to the interaction between computers and human language. Understanding, processing, and responding.

Recurrent neural network

Processing of texts of variable length (1980)

Transformer

Modeling of relationships between all words in a sentence (2017)

KÜNSTLICH INTELLIGENT

IT'S ALL ABOUT PREDICTING THE NEXT WORD

Its raining cats and dogs

dogs = 9711%



I traveled from Berlin by ICE train for 4 hours headi



2%

5%

9%

gprob on 1 tokens
y covered in top 5 logits)



Munich = 71.62%

N = 5.63%

Dresden = 3.90%

St = 3.26%

Frankfurt = 2.43%

Total: -0.33 logprob on 1 tokens
(86.84% probability covered in top 5 logits)



ckdata.de

KÜNSTLICH INTELLIGENT

AI HANDLES NUMBERS, NOT WORDS

What is the capital of France?

'[-.0.004411888774484396, 0.013418950140476227, -0.015592165291309357, -0.0025757837574929, 0.0013713511871173978, 0.004490438848733902, -0.03477144613862038, -0.024441400847434998, 0.005383944604545832, -0.009583094157278538, -0.0012928012292832136, 0.01716255583763123, -0.011461746878921986, 0.026104766875505447, -0.009308169595897198, 0.024664685130119324, 0.023368610069155693, 0.02289731055498123, 0.027701949700713158, -0.012417438440024853, -0.01843305490911007, 0.023512618616223335, 0.01570999063551426, -0.009491452015936375, -0.03416923061013222, -0.01150756794593067, 0.009642006829380089, -0.02462540939450264, 0.001659367699176073, -0.004513349384069443, 0.004860278218984604, -0.004853732418268919, -0.021444136276841164, -0.007226595655083656, -0.013432041741907597, -0.001983386231586337, -0.010970809496939182, 0.0036853018682450056, 0.024389760568737984, 0.009871110320091248, 0.002554509788751602, -0.011959230527281761, 0.025031251832842827, -0.0012183424551039934, 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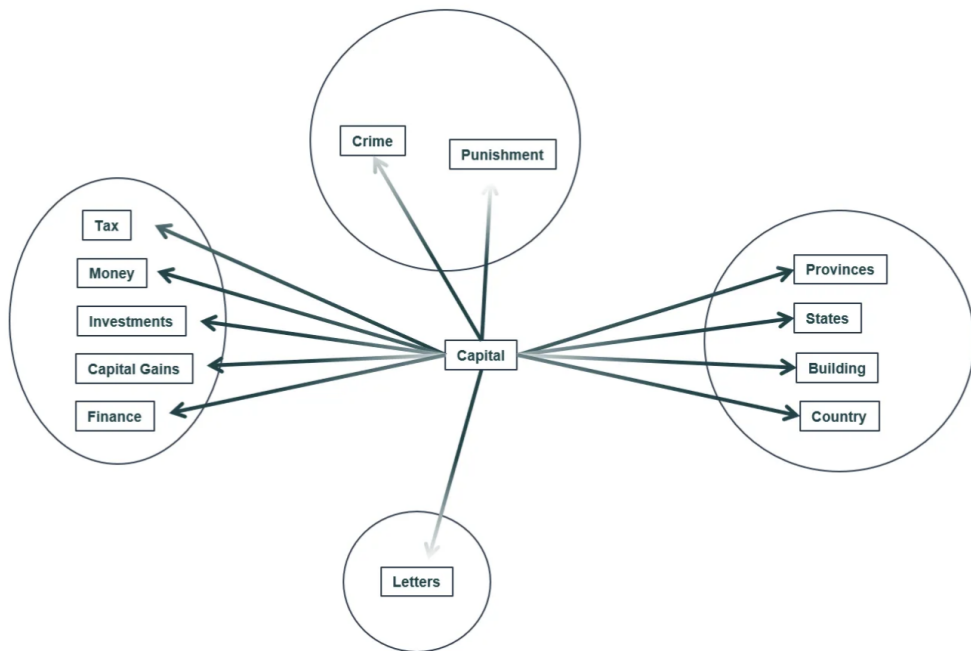


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KÜNSTLICH INTELLIGENT

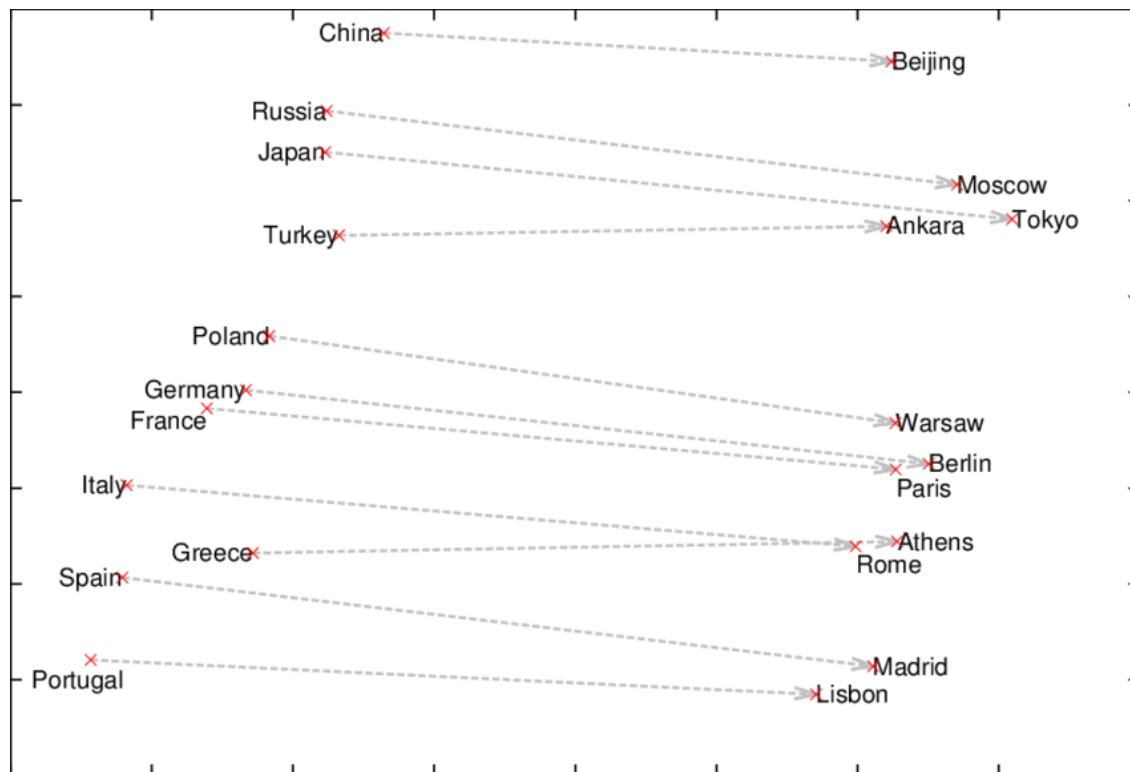
IT'S THE SEMANTIC

You can recognize a word by the company it keeps
(Firth, J.R. 1957).



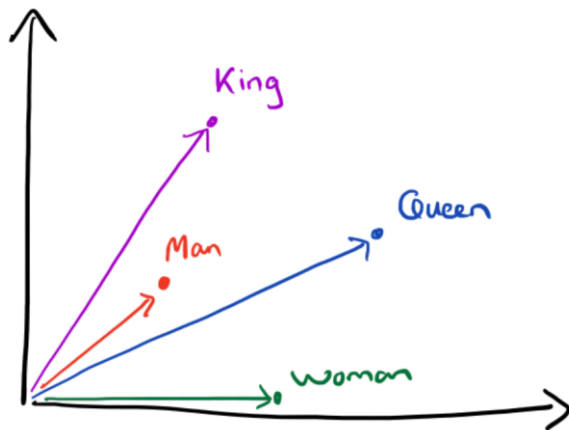
KÜNSTLICH INTELLIGENT

WORD EMBEDDING

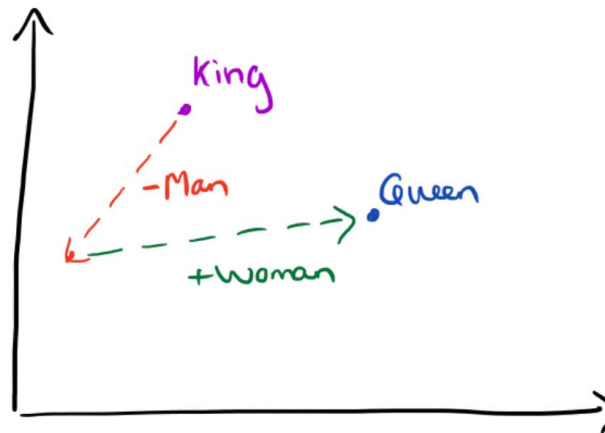


KÜNSTLICH INTELLIGENT

WORD EMBEDDING



Word
Vectors



Vector
Composition